HAZBOT / MODULAR MOBILE MANIPULATION

Physical interfaces and onboard compute

01 VISION & ONBOARD ML

Direct CSI ribbon

RGB image stream

Inference runs on Pi

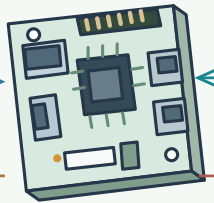
Pi Camera V2

INT8 detector · local TFLite

Task-class output → candidate target

RASPBERRY PI

Central edge computer



Native vision / ML

Range association

Navigation / mission I/O

ROS 2 + serial interfaces

02 RANGING & NAVIGATION

Scan packets

USB serial

Sensor controller

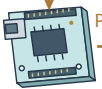
Built-in ranging MCU

YDLidar X4

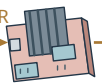
2-D ranges → clearance + target distance

03 WHEEL DRIVE

USB serial · 9600 baud



PWM + DIR



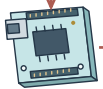
Power

Drive Arduino **L298N driver** **DC wheel pair**

Differential mobility; separate motor controller

04 ROBOTIC ARM & GRIPPER

USB serial · 115200 baud



Servo pulses

Arm Arduino

3 axes + gripper

B / S / A / G → four servo channels

Colored links denote physical interfaces; symbols are schematic, not to scale.